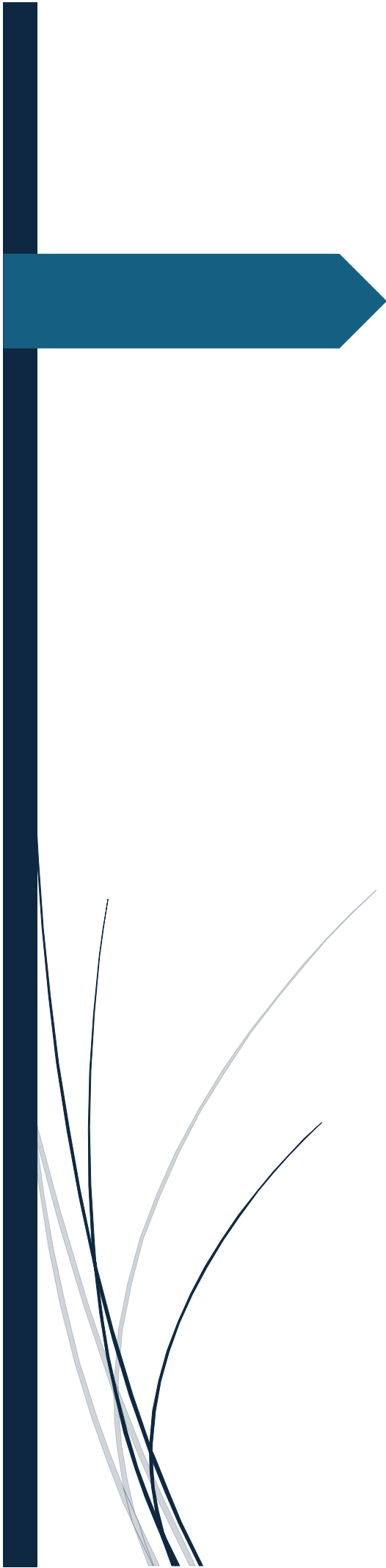




ST2187 Coursework

Business Analytics, Applied Model & Prediction

Word Count: 1493

Executive Summary

This analysis leverages Tableau's data visualisation tools to extract key business insights and present findings to senior management. We begin with a global overview, analysing all countries to identify trends and performance gaps. From this broad perspective, we narrow our focus to the United States, where profitability, despite high sales and profits, remains relatively low.

To understand the underlying issues, we conduct a detailed analysis across three dashboards, pinpointing critical problem areas within the United States, with a particular focus on Ohio, which records highly negative profitability. By examining sales, profit trends, discounting strategies, and shipping costs, we uncover the key drivers of financial underperformance in the state.

In the final dashboard, we propose a strategic solution to improve profitability by optimizing shipping costs and discounting practices. This solution is tested using four years of historical data to assess its effectiveness. Our findings demonstrate that implementing these adjustments would lead to a significant improvement in overall profitability, providing a clear and actionable path for enhancing financial performance.

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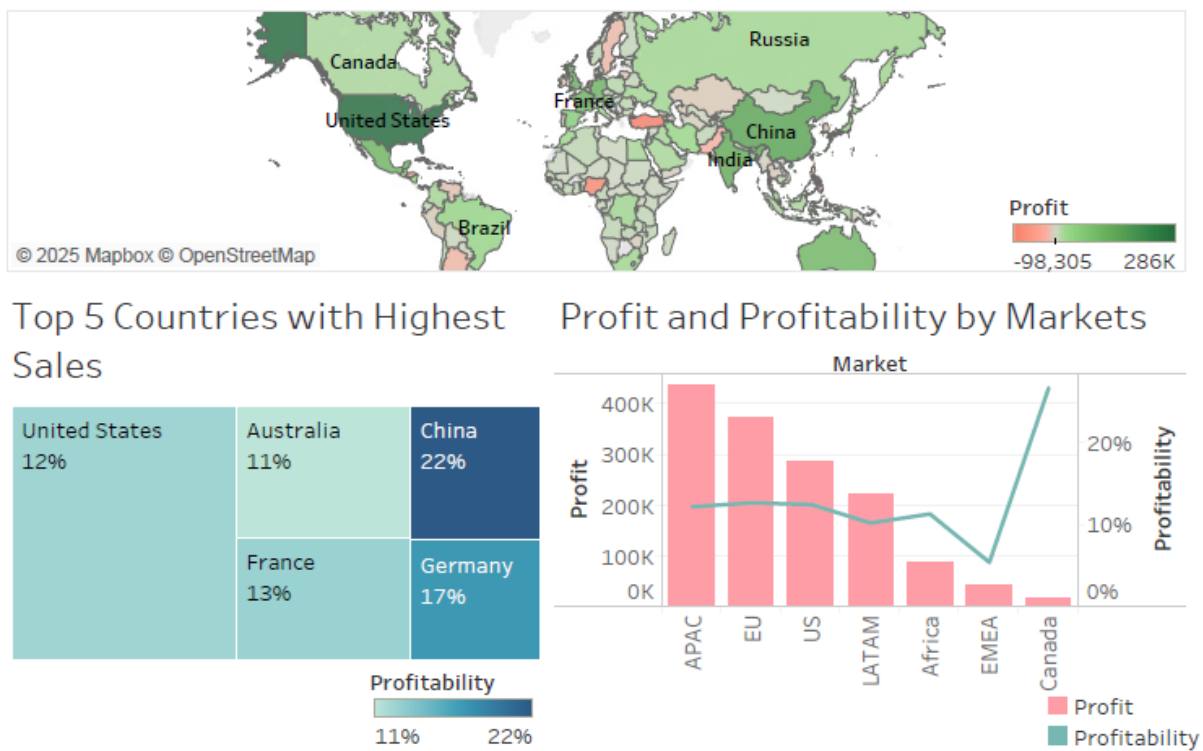
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Introduction

We are tasked to extract commercially important insights from the given dataset. I have chosen to focus on the United States (U.S.), specifically the state of Ohio, where we will conduct a deeper analysis of the challenges affecting profitability. By identifying key factors contributing to these issues, we can propose strategic improvements to optimize financial performance in the region.

Dashboard 1

Profit by Countries



This dashboard provides an in-depth comparison of sales, profit, and profitability in the U.S. relative to other countries worldwide. It features visualisations such as profit distribution by country, the top five countries with the highest sales, and an analysis of profit and profitability across different markets.

Figure 1.1 Profit by Countries

We begin by examining the total profits for each country. To visualise this, I have plotted a world map where the colour gradient represents the total profits by country. As shown in the diagram, the U.S. stands out with the highest total profits, amounting to 286,295, surpassing all other countries.

Figure 1.2 Top 5 Countries with Highest Sales

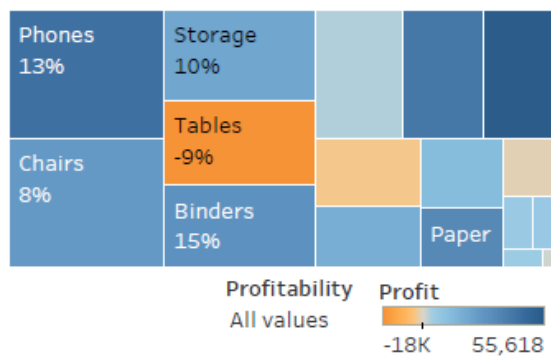
Next, we analyse the top five countries with the highest sales. In this visualisation, the size represents total sales, while the colour gradient and accompanying values indicate profitability. The U.S. leads significantly, with total sales of 2,296,487—more than double that of Australia, the second-highest country. However, despite ranking first in both total sales and profits, the U.S. has a relatively low profit margin of only 12%. In contrast, countries like China and Germany demonstrate higher profitability, with margins of 22% and 17%, respectively. This suggests that while the U.S. excels in generating revenue, there may be opportunities to improve efficiency and maximize profitability.

Figure 1.3 Profit and Profitability by Markets

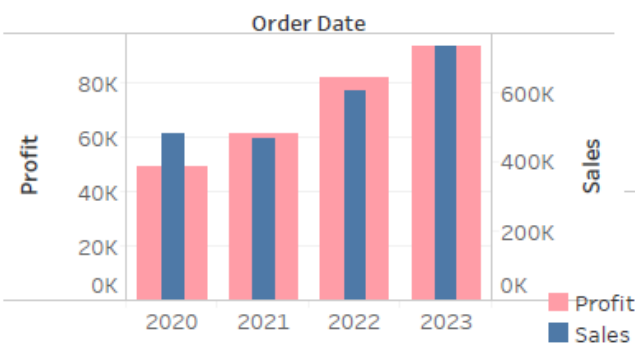
In this diagram, we compare the profit and profitability of the U.S. to global markets. The bar graphs represent profit, while the line graph illustrates profitability. Notably, the U.S. ranks third in total sales, which is impressive given that we are comparing a single country against entire markets. However, despite its high profits, the country's profitability remains relatively low. Therefore, in the following dashboards, we will take a deeper dive into the U.S. to identify potential issues and explore strategies to enhance profitability.

Dashboard 2

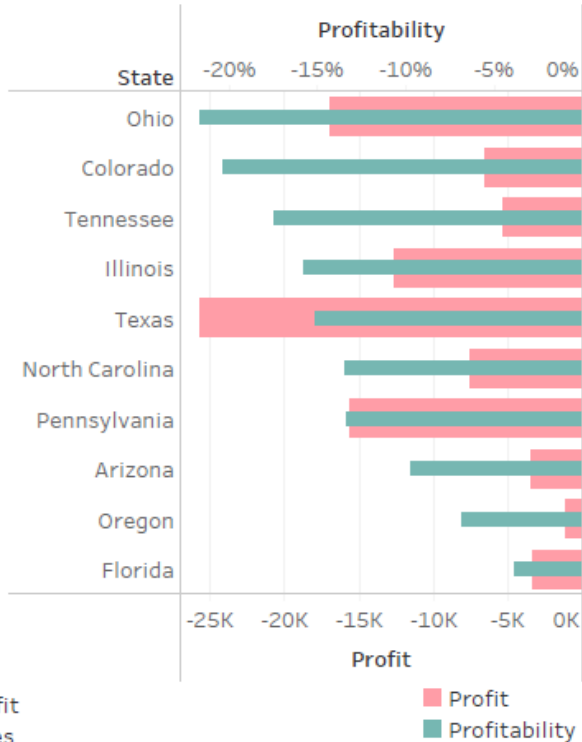
Sales & Profits by Product in US



Profit and Sales per Year in US



US States with Negative Profits



This dashboard provides an overview of the U.S., aiming to identify potential issues affecting profitability. The visualisations include sales and profits by product within the U.S., profit and sales trends over the years, and a breakdown of U.S. states with negative profits.

Figure 2.1 Sales & Profits by Products in US

This diagram provides an analysis of sales, profits, and profitability for each product in the U.S. The size of each section represents total sales, the colour gradient indicates total profit, and the percentage values denote profitability. The majority of products sold in the U.S. exhibit relatively high profitability. However, a few items fall into the negative profitability range, such as tables, which have a profitability of -9%. Given that only a small number of products experience such losses, the impact is offset by the strong performance of other items. Therefore, this is not a primary issue we will focus on for improvement.

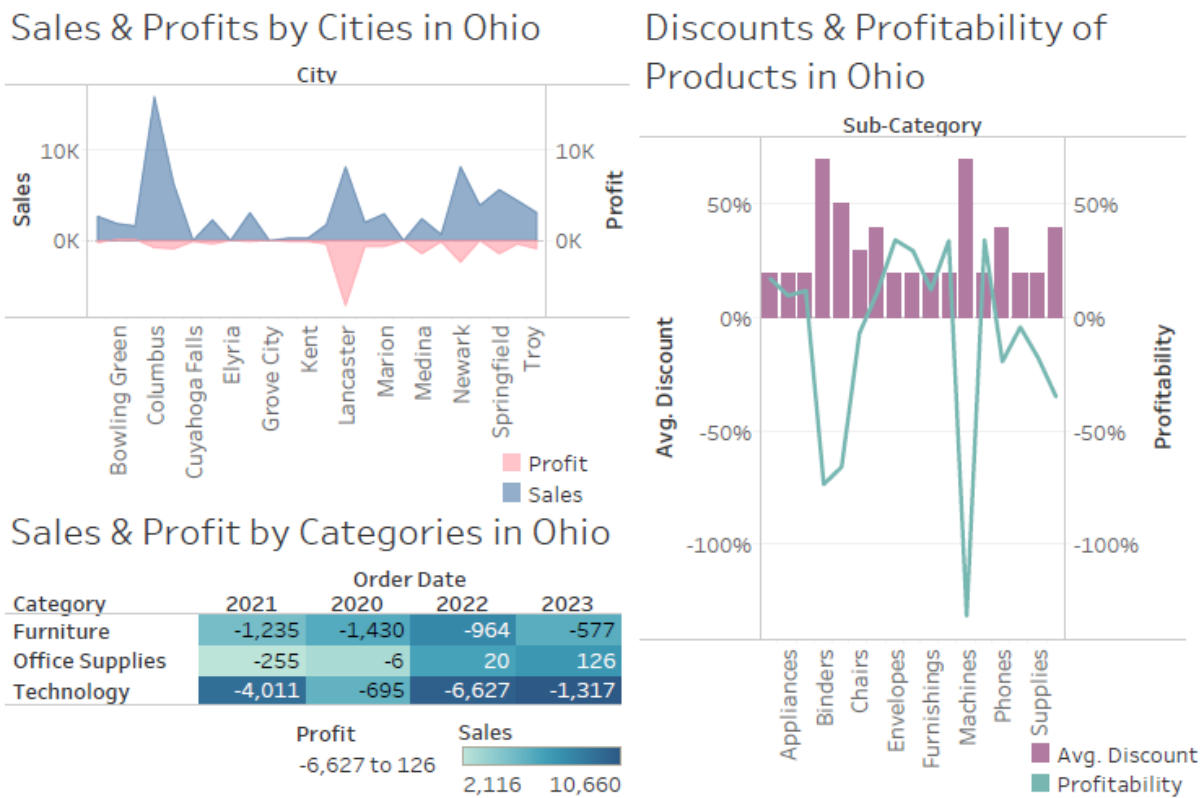
Figure 2.2 Profit and Sales per Year in US

This diagram illustrates the profit and sales trends in the U.S. over the years. The thicker pink bars represent profit, while the thinner blue bars represent sales. As shown, both profit and sales have exhibited a steady upward trend over time. This indicates that the relatively low profitability in the U.S. is not attributed to fluctuations in any specific year but rather to underlying structural factors that require further analysis.

Figure 2.3 US States with Negative Profits

This diagram highlights the U.S. states with negative profits. The thicker pink bars represent profit, while the thinner turquoise bars represent profitability. Since profitability is calculated as $\frac{Profit}{Sales}$, all these states exhibit negative profitability due to their negative profits. Notably, Ohio stands out with the lowest profitability and the second most negative profit. This suggests that Ohio may be a significant factor contributing to the overall low profitability of the U.S. Therefore, we will focus on Ohio to identify the underlying issues and explore potential solutions in the following dashboards.

Dashboard 3



In this dashboard, we take a deeper dive into Ohio to analyse the factors contributing to its extremely low profitability. The visualisations in this section include sales and profit by city, sales and profit by categories, and the relationship between discounts and profitability in Ohio.

Figure 3.1 Sales and Profits by Cities in Ohio

This diagram illustrates the sales and profits of each city in Ohio. The blue area represents sales, while the pink area represents profit. As shown, with the exception of a few cities, most cities in Ohio are experiencing negative profits, including the top three cities with the highest sales—Columbus, Lancaster, and Newark. This indicates that the low profitability is not isolated to a specific city but is a widespread issue affecting the entire state.

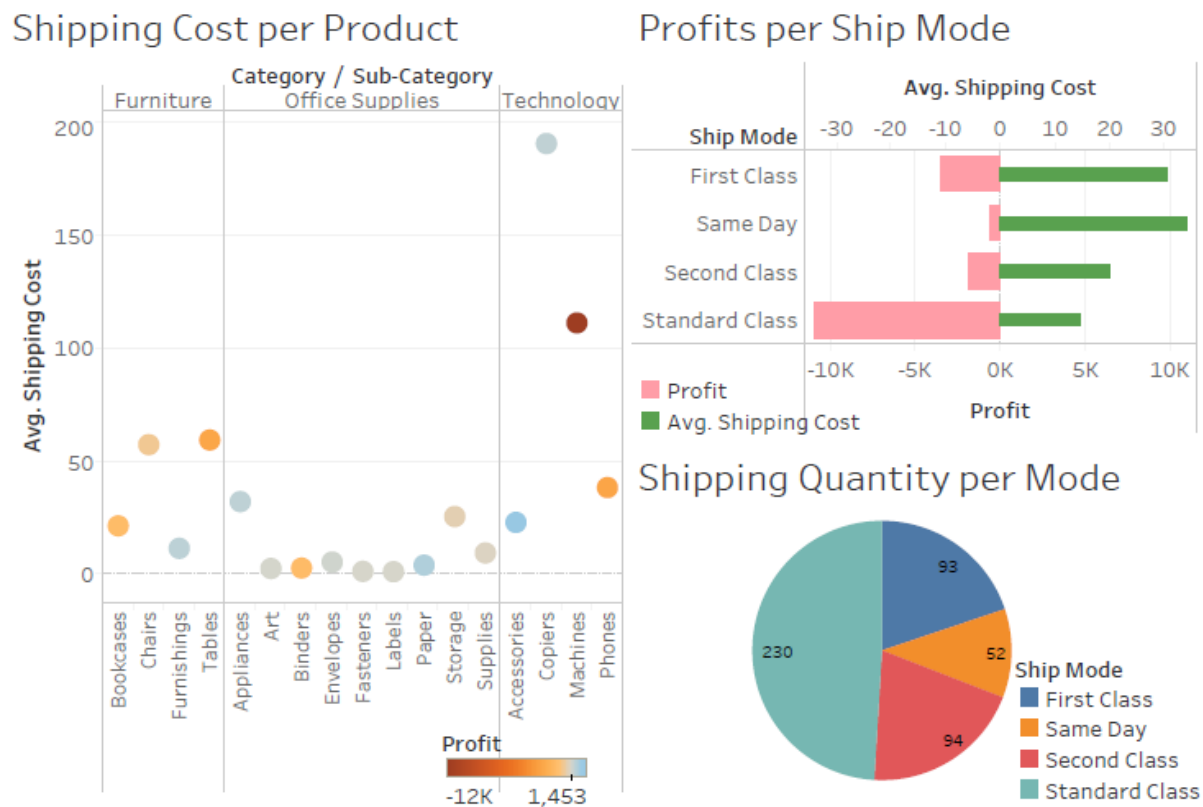
Figure 3.2 Sales and Profit by Categories in Ohio

This diagram shows the sales and profits for each category over the years in Ohio. The colour gradient represents sales, while the values within each box indicate profits. As seen, nearly every category record negative profits across all years, with the "Technology" category generating the highest negative profit. A notable trend is that as sales increase, profits tend to decrease. This relationship highlights a significant factor contributing to Ohio's low profitability, suggesting that higher sales may not be translating into proportional profit gains.

Figure 3.3 Discounts and Profitability of Products in Ohio

This diagram examines the average discounts and profitability for each product in Ohio. The bar plot represents the average discount, while the line graph illustrates the profitability for each product. As shown, most products have relatively high average discounts, and the majority exhibit negative profitability. Notably, the products with the highest average discounts—Machines and Binders—both offer a 70% discount, yet they also record the most negative profitability, with figures of -131% and -73%, respectively. These factors contribute significantly to Ohio's overall negative profitability, which will be addressed in more detail in the following dashboards.

Dashboard 4



This dashboard explores the impact of shipping on profitability in Ohio. The visualisations include shipping costs per product, profits by shipping mode, and shipping quantity by mode.

Figure 4.1 Shipping Cost per Product

This diagram analyses the average shipping cost for each product. Each point on the graph represents the average shipping cost per product, with the colour gradient indicating the associated profit for each product. As shown, office supplies generally have the lowest shipping costs, while products like copiers and machines incur the highest shipping costs. Notably, machines also generate the most negative profit, with a loss of -11,771. This aligns with previous diagrams, reinforcing that machines are a significant contributor to Ohio's negative profitability.

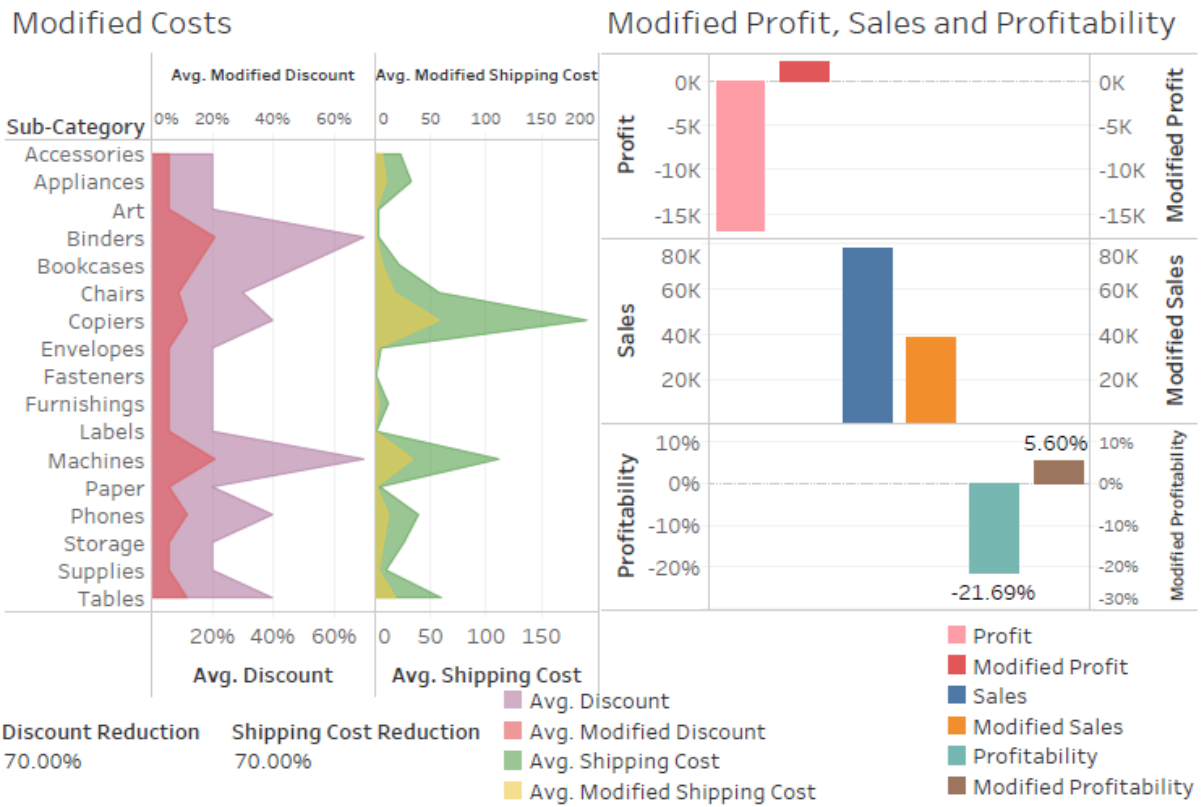
Figure 4.2 Profits per Ship Mode

This diagram examines profits and average shipping costs by shipping mode. The thicker pink bars represent profits, while the thinner green bars show the average shipping cost for each mode. As seen, all four shipping modes result in a loss. Notably, the "Same Day" shipping mode, despite having the highest average shipping cost, generates the least negative profit compared to the other modes.

Figure 4.3 Shipping Quantity per Ship Mode

This diagram presents a pie chart illustrating the shipping quantity by mode. As shown, the number of shipments using the "Same Day" shipping mode is significantly lower compared to the other three modes. This supports the previous diagram, where "Same Day" shipping, despite having a higher shipping cost, results in a smaller loss due to its lower shipping volume. Fewer shipments help mitigate the overall negative impact on profitability when compared to the higher-volume modes.

Dashboard 5



In this dashboard, we present a forward-looking analysis that explores how reducing average shipping costs and average discounts in Ohio can contribute to an increase in profitability.

Figure 5.1 Modified Costs

This diagram illustrates the impact of a suggested 70% reduction in both shipping costs and average discounts. Given the high shipping costs and average discounts highlighted in previous dashboards, this substantial reduction is necessary to improve profitability. The diagram shows a significant drop in both factors, which is expected to drive a notable increase in profitability for Ohio. The potential outcomes of these changes will be explored in the following diagram.

Figure 5.2 Modified Profit, Sales and Profitability

This diagram demonstrates the expected changes in profits, sales, and profitability following a 70% reduction in shipping costs and discounts. To understand this, we start by examining the profitability formula: $Profitability = \frac{Profit}{Sales}$. As this formula indicates, profitability is directly related to profit and inversely related to sales. The proposed reduction in shipping costs and discounts will lead to a significant increase in profit while causing a decrease in sales, as shown in the graph.

Using the adjusted values for shipping costs and discounts, I calculated the expected cost and substituted it into the formula, $Profit = 1 - \frac{Cost}{Sales}$, to determine the anticipated profit and sales. Once these values were obtained, I substituted them back into the profitability formula to estimate the expected profitability. After the reduction, the expected profitability is projected to increase by approximately 27%.

Conclusion

In conclusion, the analysis highlights that Ohio's low profitability is primarily driven by high shipping costs and discounts. By implementing a 70% reduction in both shipping costs and discounts, significant improvements in profitability are expected. This reduction will boost profits, cut sales, and enhance overall profitability, with an estimated 27% increase. Addressing these key factors will be crucial in improving Ohio's financial performance and aligning it with more profitable markets.